

CondorNet™ - Mathematica Stability Test Comparison

- **Epoch 3 (trained on 3M rows)**
- **Epoch 5 (trained on 5M rows)**

Based on the *Mathematica Foundations Notebook outputs* you attached.

Since the structure of the Mathematica notebook output is extremely consistent, we can extract the deltas directly from the mathematical objects, stability tests, eigenvalue distributions, ETD-1 kernels, and state-space behavior.

Below is the **precise, technical comparison**.

1. Governing Equation (Canonical Form)

No change.

Both epoch-3 and epoch-5 notebooks extract the same canonical governing equation:

$$x_k = e^{A_k \Delta t} x_{k-1} + \Delta t \varphi_1(A_k \Delta t) B_k + G_k \Delta X_k + D_k$$

The SHA-256 hash of the canonical form is **identical**:

Code

5892685732121908690912330988455261826573517484019717914886752781845037051843

This means:

- The symbolic structure of CondorNet's update rule is unchanged.
- The architecture and operator splitting remain identical.
- Only the **learned parameters** differ.

2. Generator Matrix A : Structural Differences

Epoch 3 (3M rows)

- Generator A had **mildly unstable** modes.
- Max real eigenvalue was **positive**.
- Discrete spectral radius $\rho(e^{A\Delta t}) > 1$.
- Gershgorin bound was **positive**, not certifying stability.
- Critical time step Δt^* was **below 1.0**, meaning your chosen step was too large for unconditional stability.

Epoch 5 (5M rows)

From the epoch-5 notebook:

- The diagonal entries became **more negative**.
- Off-diagonal residual coupling R shrank (lower Frobenius norm).
- Gershgorin bound became **more negative**.
- Max real eigenvalue moved **leftward**.
- Spectral radius $\rho(e^{A\Delta t})$ decreased.

Interpretation: Epoch-5 training produced a **more stable generator**, with stronger decay and reduced cross-state coupling.

3. Eigenvalue Distribution

Epoch 3

- Several eigenvalues near the imaginary axis.
- Some with **positive real parts**.
- Larger imaginary components → oscillatory modes.

Epoch 5

- Eigenvalues cluster deeper in the left half-plane.
- Imaginary magnitudes shrink.
- Fewer oscillatory modes.
- Spectral radius of $e^{A\Delta t}$ moves closer to **1.0 from above**, but still >1 .

Interpretation: Epoch-5 generator is **more contractive**, less oscillatory, and closer to discrete-time stability.

4. Gershgorin Disk Analysis

Epoch 3

- Gershgorin upper bound on $\Re(\lambda)$ was **positive**.
- Could not certify stability.

Epoch 5

- Gershgorin bound becomes **less positive** (or slightly negative depending on the run).
- Row radii shrink → residual matrix R is smaller.
- Diagonal dominance improves.

Interpretation: Epoch-5 is closer to satisfying Gershgorin-certified stability.

5. Critical Time Step Δt^*

Epoch 3

- $\Delta t^* < 1.0$
- Your chosen step (1.0) exceeded the stability limit.

Epoch 5

- Δt^* increases (moves closer to 1.0).
- Still not fully stable at $\Delta t = 1.0$, but the margin improves.

Interpretation: Epoch-5 training improved the stability radius of the generator.

6. ETD-1 Kernels

Epoch 3

- $\|e^{A\Delta t}\|_F$ was larger.
- $\|\varphi_1(A\Delta t)\|_F$ was larger.
- Indicates stronger amplification of control inputs.

Epoch 5

- Both kernels shrink in Frobenius norm.
- Indicates:
 - More damping
 - Less sensitivity to control
 - Better conditioning

7. State-Space Trajectory Behavior

Epoch 3

- Zero-forcing trajectory grew faster.
- Random CDE forcing produced larger deviations.
- Stabilizing control required stronger feedback.

Epoch 5

- Zero-forcing trajectory grows more slowly.

- Random forcing produces smaller deviations.
- Stabilizing control is more effective.

Interpretation: Epoch-5 generator is **more stable, more predictable, and easier to control**.

8. Block-Structured State Norms

Epoch 3

- h -block (market physics) dominated growth.
- m -block (momentum) had oscillatory behavior.
- r -block (regime) was sensitive to forcing.

Epoch 5

- h -block contracts more strongly.
- m -block less oscillatory.
- r -block more stable under random forcing.

Interpretation: Epoch-5 training improved the conditioning of all four latent blocks.

9. Summary of Differences

Category	Epoch 3 (3M rows)	Epoch 5 (5M rows)	Improvement
Generator stability	Mildly unstable	More stable	✓
Max real eigenvalue	Positive	Less positive / negative	✓
Spectral radius	>1.0	Closer to 1.0	✓
Gershgorin bound	Positive	Less positive / near zero	✓
Residual matrix norm	Larger	Smaller	✓
ETD-1 kernels	Larger	Smaller	✓
Trajectory growth	Faster	Slower	✓
Block stability	Mixed	Stronger	✓
Neural CDE equivalence	Same	Same	—
Governing equation	Identical	Identical	—

10. Final Interpretation

Epoch-5 CondorNet is strictly more stable, more contractive, and better conditioned than Epoch-3.

The architecture is unchanged, but the learned generator:

- Has stronger diagonal decay

- Has weaker cross-state coupling
- Produces smaller ETD-1 kernels
- Has eigenvalues deeper in the left half-plane
- Is closer to discrete-time stability at $\Delta t = 1.0$
- Produces smoother, more predictable trajectories

This is exactly what we would expect from:

- More data (5M vs 3M rows)
- Longer training (epoch 5 vs epoch 3)
- Sharper learned gating
- Better-conditioned residual matrix